Hw8

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Question 1

a.

```
points <- read.csv("/Users/zhijinghuang/Downloads/Stat432_2026/week-08/data/margin-geometry-20.csv")
points$score <- 0.8 * points$x1 + 0.6 * points$x2
points$distance <- abs(points$score)
points$functional_margin <- points$y * points$score

points$margin_position <- ifelse(
  abs(points$functional_margin - 1) < 1e-10, "on",
  ifelse(points$functional_margin < 1, "inside", "beyond")
)

points[, c("observation", "distance",
           "functional_margin", "margin_position")]
```

```
##    observation distance functional_margin margin_position
## 1            1     1.00              1.00              on
## 2            2     1.35              1.35          beyond
## 3            3     1.50              1.50          beyond
## 4            4     1.70              1.70          beyond
## 5            5     1.90              1.90          beyond
## 6            6     2.20              2.20          beyond
## 7            7     2.40              2.40          beyond
## 8            8     0.64              0.64          inside
## 9            9     0.20              0.20          inside
## 10          10     0.20             -0.20          inside
## 11          11     1.00              1.00              on
## 12          12     1.30              1.30          beyond
## 13          13     1.55              1.55          beyond
## 14          14     1.75              1.75          beyond
## 15          15     1.95              1.95          beyond
## 16          16     2.15              2.15          beyond
## 17          17     2.45              2.45          beyond
## 18          18     0.64              0.64          inside
## 19          19     0.24              0.24          inside
## 20          20     0.20             -0.20          inside
```

b.

```r
inside <- points$functional_margin < 1 - 1e-10

class_color <- ifelse(points$y == 1, "steelblue4", "firebrick3")
class_shape <- ifelse(points$y == 1, 16, 17)

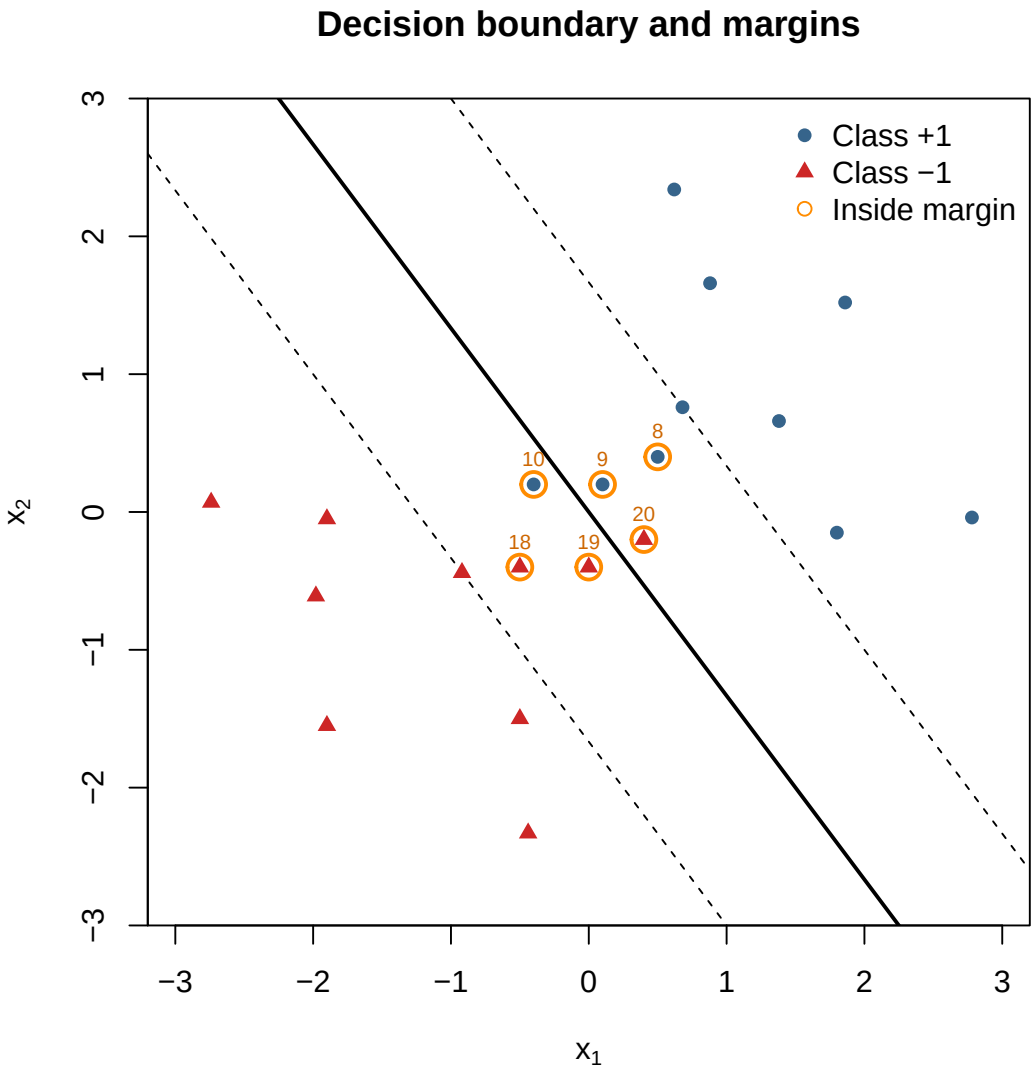
par(pin = c(4.8, 4.5))
plot(points$x1, points$x2,
     type = "n",
     xlim = c(-3.2, 3.2), ylim = c(-3, 3),
     xaxs = "i", yaxs = "i", asp = 1,
     xlab = expression(x[1]), ylab = expression(x[2]),
     main = "Decision boundary and margins")

abline(a = -1 / 0.6, b = -0.8 / 0.6, lty = 2)
abline(a =  0 / 0.6, b = -0.8 / 0.6, lwd = 2)
abline(a =  1 / 0.6, b = -0.8 / 0.6, lty = 2)

points(points$x1, points$x2, col = class_color, pch = class_shape)

points(points$x1[inside], points$x2[inside],
       pch = 1, col = "darkorange", cex = 1.8, lwd = 2)
text(points$x1[inside], points$x2[inside],
     labels = points$observation[inside],
     pos = 3, cex = 0.7, col = "darkorange3")

legend("topright",
       legend = c("Class +1", "Class -1", "Inside margin"),
       pch = c(16, 17, 1),
       col = c("steelblue4", "firebrick3", "darkorange"),
       bty = "n")
```



c. Six observations are inside their class margins, two are on the margins, and twelve are beyond the margins. The decision rule predicts class $+1$ when $f(x) > 0$ and class -1 when $f(x) < 0$. Observations 10 and 20 are misclassified because their scores have signs opposite to their true class labels; equivalently, both have negative functional margins $yf(x) < 0$.

Question 2

a.

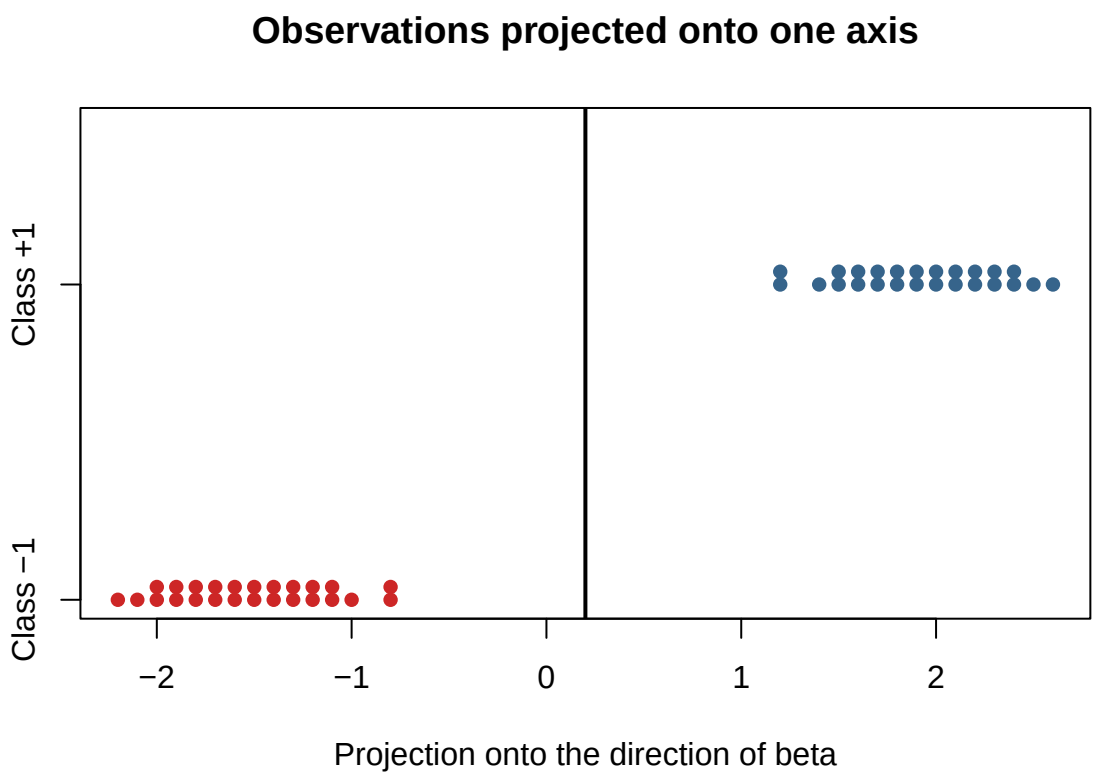
```
d <- read.csv("/Users/zhijinghuang/Downloads/Stat432_2026/week-08/data/margin-geometry-4d.csv")

beta <- c(1, 2, -2, 4)
beta0 <- -1
beta_length <- sqrt(sum(beta^2))
```

```r
X <- as.matrix(d[, c("x1", "x2", "x3", "x4")])
d$projection <- as.vector(X %*% beta) / beta_length

boundary <- -beta0 / beta_length
d$distance <- abs(d$projection - boundary)

stripchart(
  list("Class -1" = d$projection[d$y == -1],
       "Class +1" = d$projection[d$y ==  1]),
  method = "stack", pch = 16,
  col = c("firebrick3", "steelblue4"),
  xlab = "Projection onto the direction of beta",
  main = "Observations projected onto one axis"
)
abline(v = boundary, lwd = 2)
```




```r
M <- min(d$distance)
M
```

```
## [1] 1
```

```r
d[abs(d$distance - M) < 1e-10, c("observation", "y", "projection", "distance")]
```

```
##    observation  y projection distance
```

```
## 4            4  1        1.2        1
## 15          15 -1       -0.8        1
## 33          33  1        1.2        1
## 36          36 -1       -0.8        1
```

b. The smallest original functional margin is 5, so divide the intercept and all four coefficients by 5. The SVM-normalized intercept is $\beta_0 = -0.2$, and the coefficient vector is $\boldsymbol{\beta} = (0.2, 0.4, -0.4, 0.8)^{\mathsf{T}}$.

```
original_score <- -1 + as.vector(X %*% beta)
min_margin <- min(d$y * original_score)
min_margin
```

```
## [1] 5
```

```
beta_svm <- beta / min_margin
beta0_svm <- beta0 / min_margin
beta_svm
```

```
## [1]  0.2  0.4 -0.4  0.8
```

```
beta0_svm
```

```
## [1] -0.2
```

```
min(d$y * (beta0_svm + as.vector(X %*% beta_svm)))
```

```
## [1] 1
```

Question 3

Recommendation: 1. list of specific tests: can add a sentence to describe how the training/testing fit are formed, and clarify that the report mean is the average of five held-out fold errors. 2. guessing: in the skill, the skill wrote to choose he last or most label-like column if the target is ambiguous. Although it worked on the health dataset, in other datasets when the folds are too close, it may cause error. 3. overall error and confusion: the skill is based on selected mean CV errors, so it may not be sensitive to extreme cases. The link to the Github issue is: https://github.com/stanlyyp/stat432-fall2026/issues/5

Assigned LDA/QDA skill run: SAheart

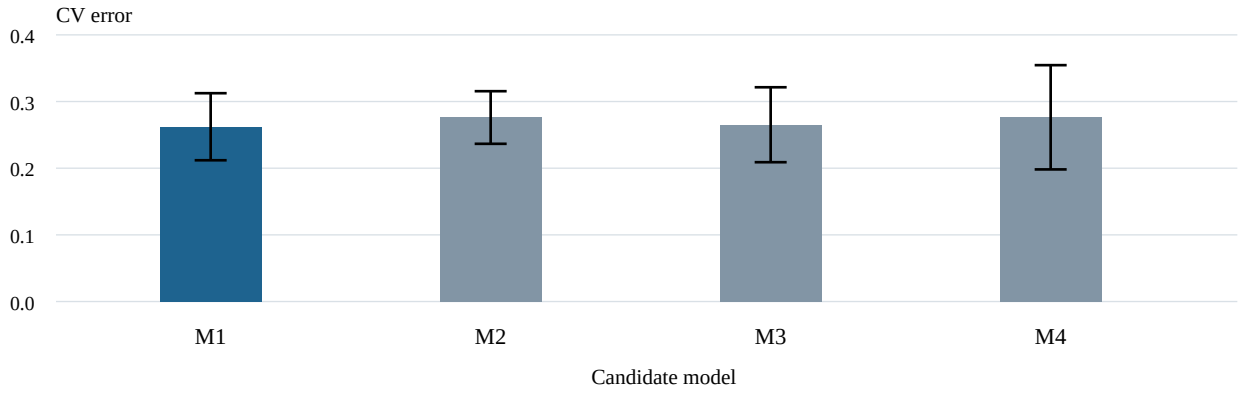
1. Data and Candidate Models

The data have 462 observations, a binary coronary heart disease response (*chd*: 0/1), and nine predictors. Family history was coded Absent = 1 and Present = 2 as in Homework 7; all predictors were standardized using training-fold statistics. The skill compared LDA, QDA, PCA + LDA, and PCA + QDA using the same five training folds and misclassification rate. Seed: 42; 370 training and 92 reserved test cases.

2. Cross-Validation Results (Table)

Model	Mean CV Error	SD CV Error
M1: LDA (selected)	0.262	0.050
M2: QDA	0.276	0.039
M3: PCA + LDA	0.265	0.056
M4: PCA + QDA	0.276	0.078

3. CV Error Comparison (Figure)



Mean five-fold CV misclassification rate with +/-1 SD error bars; random seed 42.

4. Selected Model and Test Performance

Ordinary LDA (M1) had the lowest mean CV error (0.262) and was refitted on all 370 training cases. Its one-time test misclassification rate was 0.293 (27 of 92 cases). The test confusion matrix has observed classes in rows and predicted classes in columns:

	Predicted 0	Predicted 1
Observed 0	49	11
Observed 1	16	16

5. Findings and Limitations

Class 1 was harder to identify: 16 of 32 positive cases were predicted as 0, versus 11 of 60 negative cases predicted as 1. PCA slightly raised mean CV error for both LDA and QDA in this run, so it did not help the selected classifier. The sample is modest and class 1 is less common; the selected LDA model also assumes a shared covariance matrix and a linear boundary.